

WHAT IS DYNAMIC TIME WARPING ?

```
# IT IS THE ALGORITHM USED TO MEASURE THE SIMILARITY BETWEEN THE TWO  
# TIME SERIES DATA  
# IT FINDS THE OPTIMAL ALIGNMENT BETWEEN THE SERIES .IT CAN  
# PERFORM ALIGNMENT OF NON LINEAR DATA ALSO, THAT IS SPECIALITY IN THE  
# DTW  
# IT ALLIGNS THE TWO SIGNALS BY STRECHING AND COMPRESSING, HENCE IT IS  
# CALLED WARPING.
```

WHAT IS OPTIMAL ALIGNMENT ?

```
# OPTIMAL ALIGNMENT REFERS TO THE BEST POSSIBLE ALIGNMENT THAT MATCHES  
# TWO SIGNALS BY MINIMIZING THE PREDEFINED COST OR DISTANCE FUNCTION.
```

```
pip install fastdtw
```

```
Requirement already satisfied: fastdtw in  
/opt/anaconda3/lib/python3.12/site-packages (0.3.4)  
Requirement already satisfied: numpy in  
/opt/anaconda3/lib/python3.12/site-packages (from fastdtw) (1.26.4)  
Note: you may need to restart the kernel to use updated packages.
```

```
pip install dtaidistance
```

```
Requirement already satisfied: dtaidistance in  
/opt/anaconda3/lib/python3.12/site-packages (2.3.12)  
Requirement already satisfied: numpy in  
/opt/anaconda3/lib/python3.12/site-packages (from dtaidistance)  
(1.26.4)  
Note: you may need to restart the kernel to use updated packages.
```

1)PERFORMING THE DTW DISTANCE ON THE SYNTHETIC DATA

```
import numpy as np  
import matplotlib.pyplot as plt  
from scipy.spatial.distance import euclidean  
from fastdtw import fastdtw  
  
well_log_1 = np.array([1, 2, 3, 4, 2, 3, 1])  
well_log_2 = np.array([1, 1, 2, 3, 4, 3, 2])  
  
distance, path = fastdtw(well_log_1, well_log_2, dist=lambda x, y:  
euclidean([x], [y]))  
distance, path  
  
(2.0, [(0, 0), (0, 1), (1, 2), (2, 3), (3, 4), (4, 5), (5, 5), (6,  
6)])
```

PLOTTING THE DISTANCE MATRIX OF THE TWO WELL LOG DATA BY HEAT MAPS

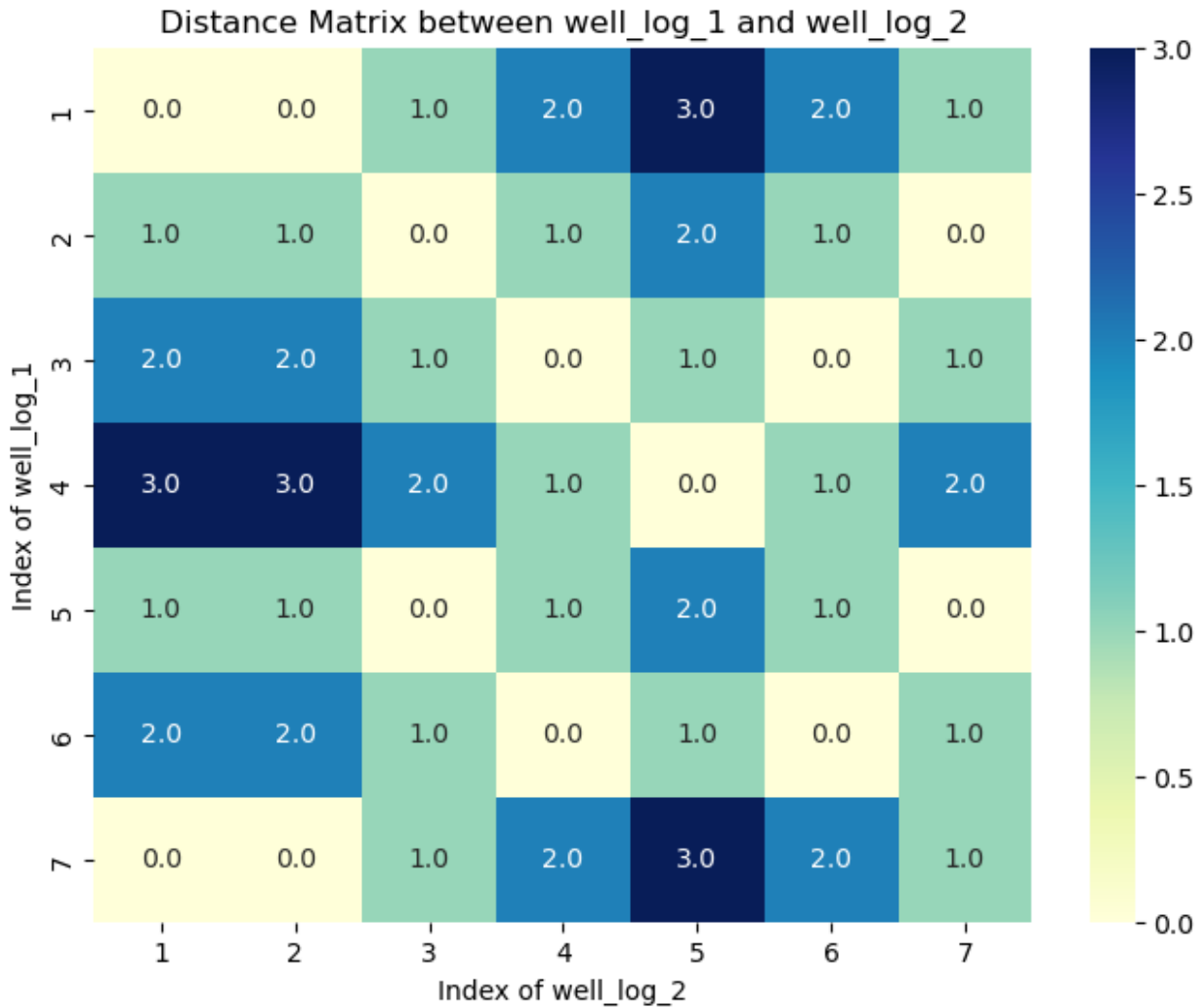
```
import numpy as np
import matplotlib.pyplot as plt
from scipy.spatial.distance import euclidean
import seaborn as sns

# Define the sequences (well logs)
well_log_1 = np.array([1, 2, 3, 4, 2, 3, 1])
well_log_2 = np.array([1, 1, 2, 3, 4, 3, 2])

# Initialize the distance matrix
distance_matrix = np.zeros((len(well_log_1), len(well_log_2)))

# Calculate the pairwise Euclidean distance
for i in range(len(well_log_1)):
    for j in range(len(well_log_2)):
        distance_matrix[i, j] = abs(well_log_1[i] - well_log_2[j]) # Euclidean for 1D

# Plot the distance matrix as a heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(distance_matrix, annot=True, fmt=".1f", cmap="YlGnBu",
            xticklabels=range(1, len(well_log_2) + 1),
            yticklabels=range(1, len(well_log_1) + 1))
plt.title("Distance Matrix between well_log_1 and well_log_2")
plt.xlabel("Index of well_log_2")
plt.ylabel("Index of well_log_1")
plt.show()
```



INTERPRETING AND VISUALIZING THE DTW DISTANCE AND OPTIMAL ALIGNMENT PATH ON THE HEAT MAP

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.spatial.distance import euclidean
import seaborn as sns
from fastdtw import fastdtw

# Define the sequences (well logs)
well_log_1 = np.array([1, 2, 3, 4, 2, 3, 1])
well_log_2 = np.array([1, 1, 2, 3, 4, 3, 2])

# Calculate DTW distance and path
distance, path = fastdtw(well_log_1, well_log_2, dist=lambda x, y:
euclidean([x], [y]))

# Initialize the distance matrix
```

```

distance_matrix = np.zeros((len(well_log_1), len(well_log_2)))

# Calculate the pairwise Euclidean distance
for i in range(len(well_log_1)):
    for j in range(len(well_log_2)):
        distance_matrix[i, j] = abs(well_log_1[i] - well_log_2[j])

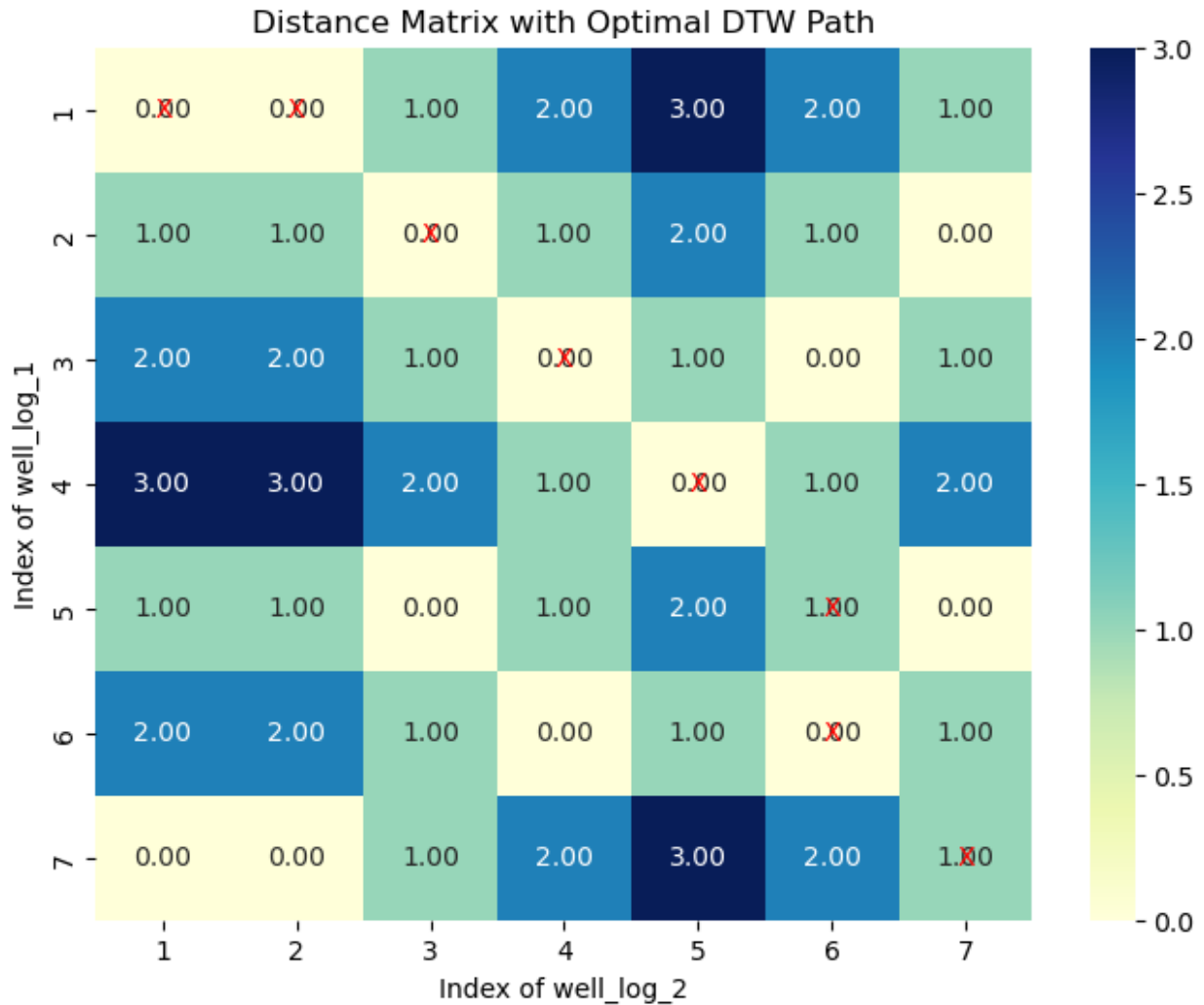
# Plot the distance matrix as a heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(distance_matrix, annot=True, fmt=".2f", cmap="YlGnBu",
            xticklabels=range(1, len(well_log_2)+1), yticklabels=range(1,
            len(well_log_1)+1))

# Overlay the optimal path on the heatmap
for (i, j) in path:
    plt.text(j+0.5, i+0.5, 'X', color='red', ha='center',
            va='center')

plt.title("Distance Matrix with Optimal DTW Path")
plt.xlabel("Index of well_log_2")
plt.ylabel("Index of well_log_1")
plt.show()

# Output DTW distance and path
print(f"DTW Distance: {distance}")
print(f"DTW Path: {path}")

```



DTW Distance: 2.0

DTW Path: [(0, 0), (0, 1), (1, 2), (2, 3), (3, 4), (4, 5), (5, 5), (6, 6)]

FOR CALCULATING THE ALIGNED WELL LOGS

```
# Create aligned sequences
aligned_well_log_1 = []
aligned_well_log_2 = []

# Using the DTW path to align the sequences
for (i, j) in path:
    aligned_well_log_1.append(well_log_1[i])
    aligned_well_log_2.append(well_log_2[j])

# Convert aligned lists to numpy arrays (optional)
aligned_well_log_1 = np.array(aligned_well_log_1)
```

```
aligned_well_log_2 = np.array(aligned_well_log_2)
```

```
print("Aligned Well Log 1:", aligned_well_log_1)
print('Orginal well log 1:',well_log_1)
print("Aligned Well Log 2:", aligned_well_log_2)
print('Orginal well log 2:',well_log_2)
```

```
Aligned Well Log 1: [1 1 2 3 4 2 3 1]
Orginal well log 1: [1 2 3 4 2 3 1]
Aligned Well Log 2: [1 1 2 3 4 3 3 2]
Orginal well log 2: [1 1 2 3 4 3 2]
```

```
np.random.normal()
```

```
1.060879489562633
```

```
import numpy as np
from fastdtw import fastdtw
from scipy.spatial.distance import euclidean
import matplotlib.pyplot as plt
depth = np.linspace(0, 100, 100)
well_log_1_1 = np.sin(depth) + np.random.normal(loc=0, scale=0.1,size
=len(depth))
well_log_2_2 = np.sin(depth + 0.5) + np.random.normal(loc=0,scale=0.1,
size=len(depth))
```

```
distance,path=fastdtw(well_log_1_1,well_log_2_2,dist=lambda x,y :
euclidean([x],[y]),radius = 1)
aligned_well_log_1_1=[well_log_1_1[i]for (i,j) in path]
aligned_well_log_2_2=[well_log_2_2[j] for (i,j) in path]
aligned_depth = np.linspace(0,100,len(aligned_well_log_1_1))
print(f"DTW distance between the two well logs: {distance}")
print(f"Alignment path : {path}")
plt.figure(figsize=(12,6))
plt.subplot(2,2,1)
plt.plot(depth,well_log_1_1,label="well_log_1",color='blue')
plt.xlabel("DEPTH")
plt.ylabel('WELL LOG SIGNAL 1')
plt.subplot(2,2,2)
plt.plot(depth,well_log_2_2,label='well_log_2',color='red')
plt.xlabel("DEPTH")
plt.ylabel('WELL LOG SIGNAL 2')
plt.subplot(2,2,3)
plt.plot(aligned_depth,aligned_well_log_1_1,label='aligned_well_log
_1',color = 'blue')
plt.xlabel('ALLIGNED WELL DEPTH')
plt.ylabel('ALLIGNED WELL LOG 1')
plt.subplot(2,2,4)
plt.plot(aligned_depth,aligned_well_log_2_2,label='aligned_well_log
```

```

_2',color = 'red')
plt.xlabel('ALLIGNED WELL DEPTH')
plt.ylabel('ALLIGNED WELL LOG 2')

```

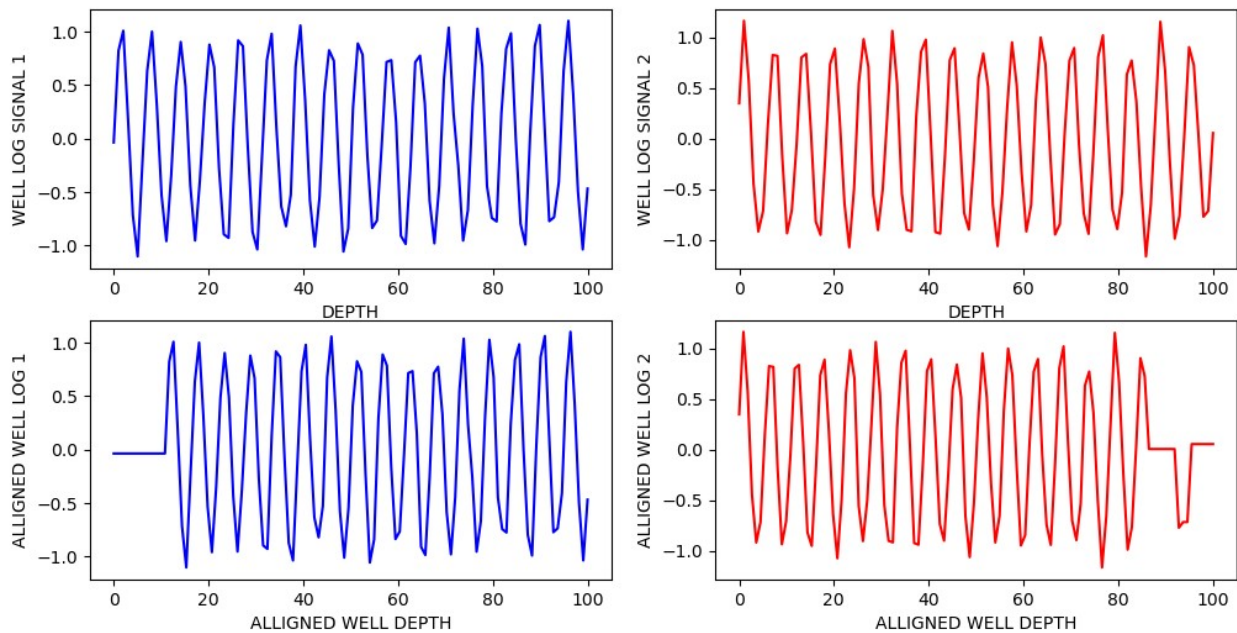
DTW distance between the two well logs: 25.689805172981405

Alignment path : [(0, 0), (0, 1), (0, 2), (0, 3), (0, 4), (0, 5), (0, 6), (0, 7), (0, 8), (0, 9), (0, 10), (0, 11), (0, 12), (1, 13), (2, 14), (3, 15), (4, 16), (5, 17), (6, 18), (7, 19), (8, 20), (9, 21), (10, 22), (11, 23), (12, 24), (13, 25), (14, 26), (15, 27), (16, 28), (17, 29), (18, 30), (19, 31), (20, 32), (21, 33), (22, 34), (23, 35), (24, 36), (25, 37), (26, 38), (27, 39), (28, 40), (29, 41), (30, 42), (31, 43), (32, 44), (33, 45), (34, 46), (35, 47), (36, 48), (37, 49), (38, 50), (39, 51), (40, 52), (41, 53), (42, 54), (43, 55), (44, 56), (45, 57), (46, 58), (47, 59), (48, 60), (49, 61), (50, 62), (51, 63), (52, 64), (53, 65), (54, 66), (55, 67), (56, 68), (57, 69), (58, 70), (59, 71), (60, 72), (61, 73), (62, 74), (63, 75), (64, 76), (65, 77), (66, 78), (67, 79), (68, 80), (69, 81), (70, 82), (71, 83), (72, 84), (73, 85), (74, 86), (75, 87), (76, 88), (77, 89), (78, 90), (79, 91), (80, 92), (81, 93), (82, 94), (83, 95), (84, 96), (85, 96), (86, 96), (87, 96), (88, 96), (89, 96), (90, 96), (91, 97), (92, 98), (93, 98), (94, 99), (95, 99), (96, 99), (97, 99), (98, 99), (99, 99)]

```

Text(0, 0.5, 'ALLIGNED WELL LOG 2')

```



```

len(aligned_well_log_1_1)

```

```

112

```

```

len(well_log_1_1)

```

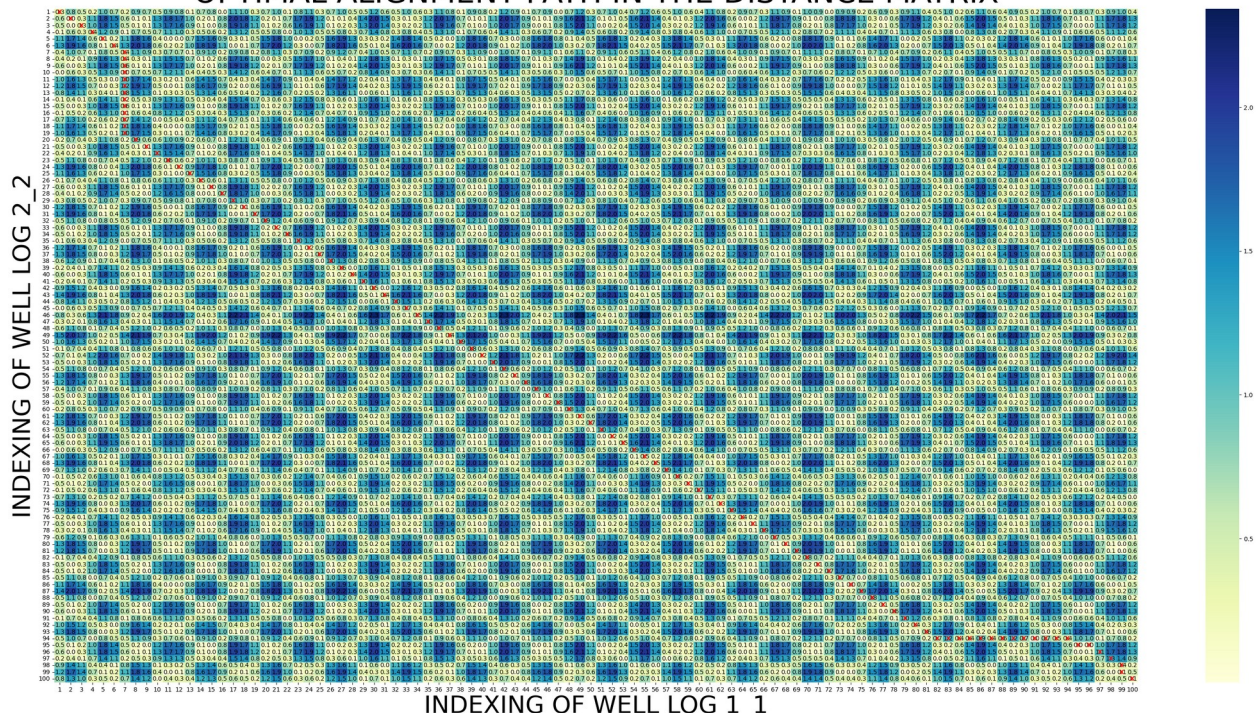
100

DISTANCE MATRIX

```
distance_matrix = np.zeros((len(well_log_1_1),len(well_log_2_2)))
for i in range(len(well_log_1_1)):
    for j in range(len(well_log_2_2)):
        distance_matrix[i,j] = euclidean([well_log_1_1[i]],
[well_log_2_2[j]])
plt.figure(figsize=(40,20))
sns.heatmap(distance_matrix,annot=True,fmt='.1f',cmap="YlGnBu",annot_k
ws={"size": 10, "color": "black"},
        cbar=True, xticklabels=range(1, len(well_log_2_2)+1),
yticklabels=range(1, len(well_log_1_1)+1))
for (i,j) in path:

plt.text(i+0.5,j+0.5,'X',va='center',ha='center',color='red',fontweigh
t='bold')
plt.xlabel('INDEXING OF WELL LOG 1_1',fontsize=40)
plt.ylabel('INDEXING OF WELL LOG 2_2',fontsize=40)
plt.title('OPTIMAL ALIGNMENT PATH IN THE DISTANCE MATRIX',fontsize=50)
plt.show()
```

OPTIMAL ALIGNMENT PATH IN THE DISTANCE MATRIX

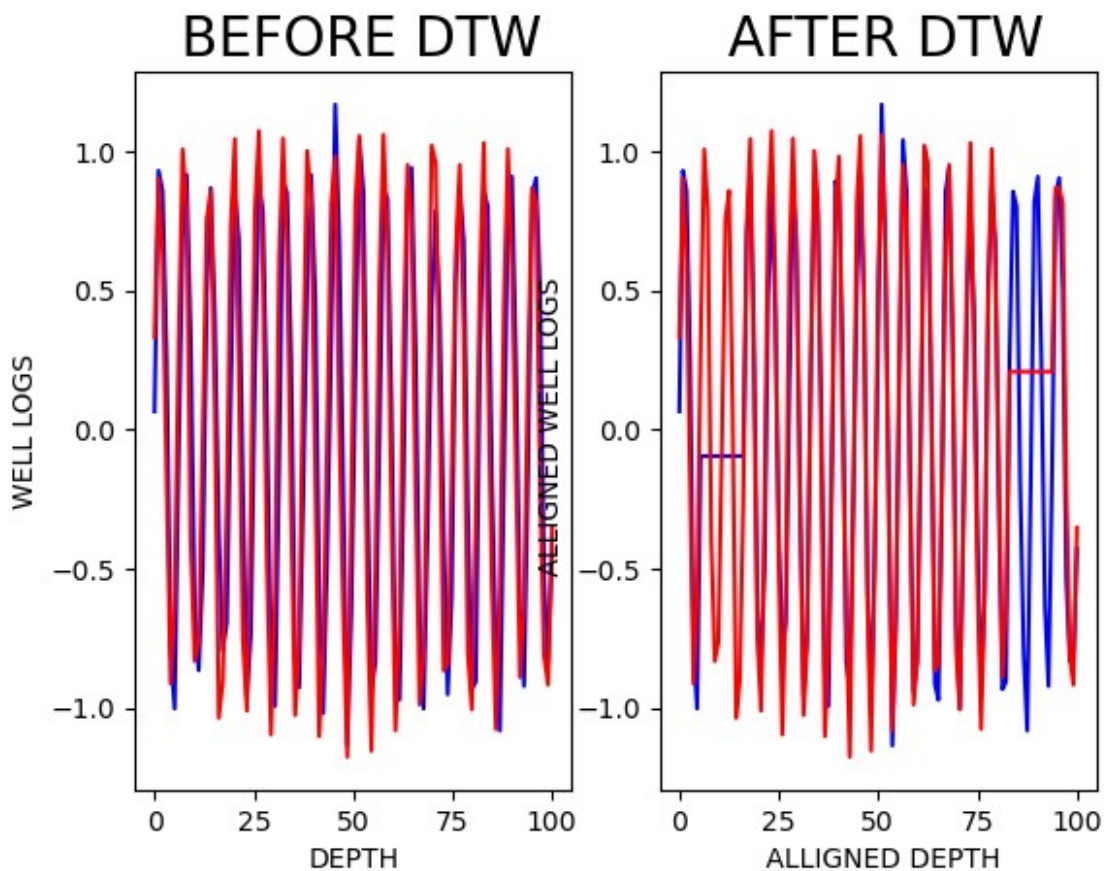


SEEING THE DIFFERENCE BETWEEN THE ALIGNMENT BEFORE AND AFTER APPLYING THE DTW

```
# BEFORE DTW
plt.subplot(1,2,1)
plt.plot(depth,well_log_1_1,label='well_log_1_1',color='blue')
plt.plot(depth,well_log_2_2,label='well_log_2_2',color='red')
plt.xlabel('DEPTH')
plt.ylabel('WELL LOGS')
plt.title(' BEFORE DTW',fontsize=20)

# AFTER DTW
plt.subplot(1,2,2)
plt.plot(aligned_depth,aligned_well_log_1_1,label='well_log_1',color='blue')
plt.plot(aligned_depth,aligned_well_log_2_2,label='well_log_2',color='red')
plt.xlabel(' ALIGNED DEPTH')
plt.ylabel(' ALIGNED WELL LOGS')
plt.title(' AFTER DTW',fontsize=20)

Text(0.5, 1.0, ' AFTER DTW')
```



```

import numpy as np
import matplotlib.pyplot as plt
from scipy.spatial.distance import euclidean
from fastdtw import fastdtw

# ORIGINAL WELL LOGS

np.random.seed(0)
depth = np.linspace(0, 100, 100)
well_log_1 = np.sin(np.linspace(0, 10, 100)) + np.random.normal(0,
0.1, 100)
well_log_2 = np.sin(np.linspace(0.5, 10.5, 100)) + np.random.normal(0,
0.1, 100)
distance,path=fastdtw(well_log_1,well_log_2,dist=lambda x,y:
euclidean([x],[y]))
print(f"DTW distance: {distance}")
print(f"path:{path}")

# ALLIGNED WELL LOGS AFTER DTW

aligned_well_log_1=[well_log_1[i] for (i,j) in path]
aligned_well_log_2=[well_log_2[j] for (i,j) in path]
aligned_depth = np.linspace(0,100,len(aligned_well_log_1))
print(aligned_well_log_1)
print(aligned_well_log_2)
plt.figure(figsize=(12,6))
plt.subplot(1,2,1)
plt.plot(depth,well_log_1,color='blue')
plt.plot(depth,well_log_2,color='red')
plt.ylabel('ORIGINAL WELL LOGS')
plt.xlabel('DEPTH')
plt.title('BEFORE DTW',fontsize=30)
plt.subplot(1,2,2)
plt.plot(aligned_depth,aligned_well_log_1,color='blue')
plt.plot(aligned_depth,aligned_well_log_2,color='red')
plt.xlabel('ALLIGNED DEPTH')
plt.ylabel('ALLIGNED WELL LOGS')
plt.title('AFTER DTW',fontsize=30)

DTW distance: 9.198101926243027
path:[(0, 0), (1, 1), (2, 1), (3, 2), (4, 2), (5, 2), (6, 3), (7, 4),
(8, 4), (9, 5), (10, 6), (10, 7), (11, 8), (12, 9), (13, 10), (14,
11), (15, 12), (16, 13), (17, 14), (18, 15), (19, 16), (20, 17), (21,
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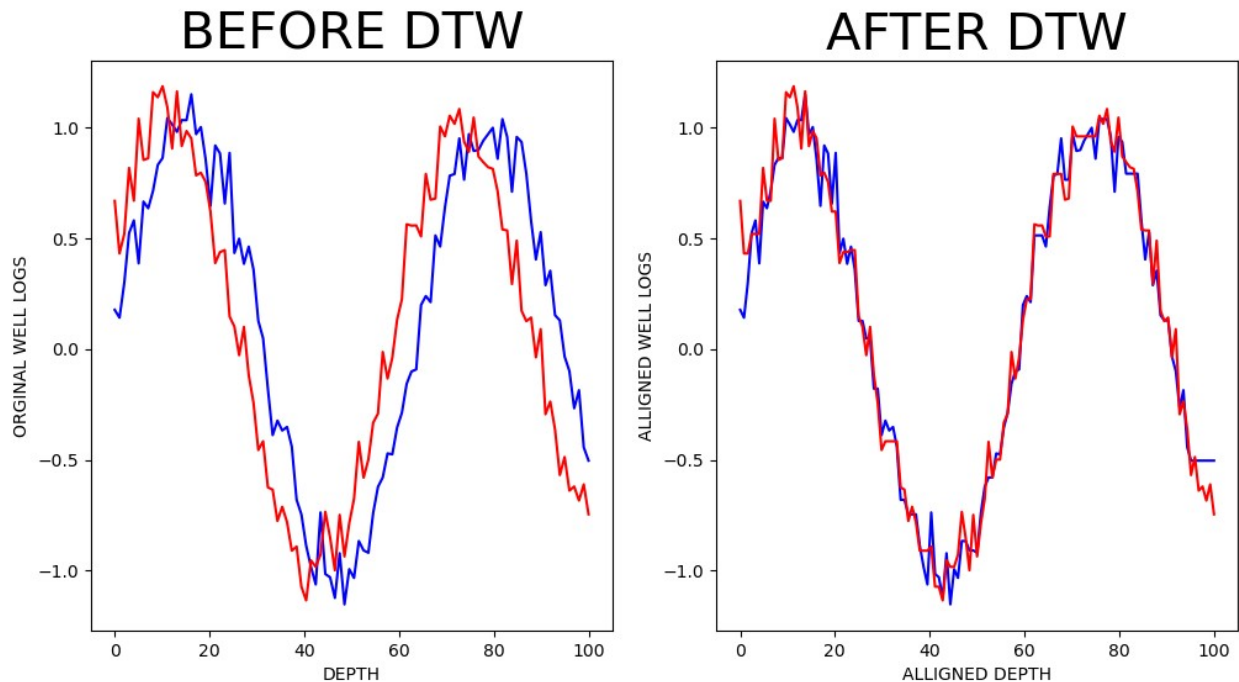
```

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1.043667799281705, 0.8680143729030183, 0.8425794698168936,
0.8198677876472541, 0.8122331235244611, 0.7118876014224444,
0.5389755293142064, 0.5343962934516926, 0.5343962934516926,
0.29260133266177524, 0.4883988023316682, 0.17226236771902576,
0.1252329727934831, 0.1416884680236603, -0.038074931602328735,
0.08922644763633283, -0.2946536841802382, -0.2372700653509952, -
0.3638195332658364, -0.568947836414139, -0.4874487895345383, -
0.6390663776770767, -0.6205279810350328, -0.684039229879024, -
0.6109358011715523, -0.7460429650280309]

Text(0.5, 1.0, 'AFTER DTW')



ON REAL TIME DATA

```
import pandas as pd
df1=pd.read_excel('data_2.xlsx')
df1.columns=[f'well_log_data{i+1}' for i in range(709)]
df1.head()
df1.shape
```

```
(2999, 709)
```

```
df1.shape
```

```
(2999, 709)
```

```
df1.head()
```

	well_log_data1	well_log_data2	well_log_data3	well_log_data4	\
0	-0.077591	0.740433	0.169373	0.599976	
1	0.122147	0.786514	0.019760	0.913315	
2	0.322952	0.751419	-0.144501	1.093750	
3	0.454788	0.678940	-0.220795	1.231918	
4	0.525742	0.605698	-0.146790	1.329193	

	well_log_data5	well_log_data6	well_log_data7	well_log_data8	\
0	1.950912	1.081314	-0.157242	1.768799	
1	1.363907	2.126541	-0.484009	0.435410	
2	0.643005	3.010864	-0.552750	-0.783768	
3	-0.048065	3.705978	-0.727310	-1.676254	
4	-0.736313	4.136887	-0.994339	-2.016373	

```

    well_log_data9 well_log_data10 ... well_log_data700
well_log_data701 \
0      -2.150650      1.593857 ...      0.056992
0.047150
1      -2.129898      2.189331 ...      0.053711
0.062408
2      -1.842194      2.485428 ...      0.050507
0.051956
3      -1.122284      2.801437 ...      0.052795
0.036545
4      -0.271301      2.626953 ...      0.055542
0.025024

    well_log_data702 well_log_data703 well_log_data704
well_log_data705 \
0      -0.056229      -0.002289      0.061722      -
0.034409
1      -0.069427      -0.008316      0.077362      -
0.046005
2      -0.004196      -0.014801      0.087662      -
0.053558
3      -0.012207      -0.023422      0.083237      -
0.056610
4      -0.078659      -0.036621      0.073929      -
0.064697

    well_log_data706 well_log_data707 well_log_data708
well_log_data709
0      -0.043564      -0.014267      -0.014267      -
0.039291
1      -0.038147      -0.018234      -0.019226      -
0.041809
2      -0.036469      -0.023041      -0.023499      -
0.059967
3      -0.034866      -0.018921      -0.032959      -
0.080109
4      -0.029602      -0.019379      -0.031204      -
0.083466

[5 rows x 709 columns]

df1.isnull().sum()*100

well_log_data1      0
well_log_data2      0
well_log_data3      0
well_log_data4      0
well_log_data5      0
..
well_log_data705    0

```

```

well_log_data706      0
well_log_data707      0
well_log_data708      0
well_log_data709      0
Length: 709, dtype: int64

well_log_data1=df1['well_log_data1']
well_log_data1
0      -0.077591
1       0.122147
2       0.322952
3       0.454788
4       0.525742
...
2994     0.926666
2995     1.022797
2996     1.166534
2997     1.302795
2998     1.410599
Name: well_log_data1, Length: 2999, dtype: float64

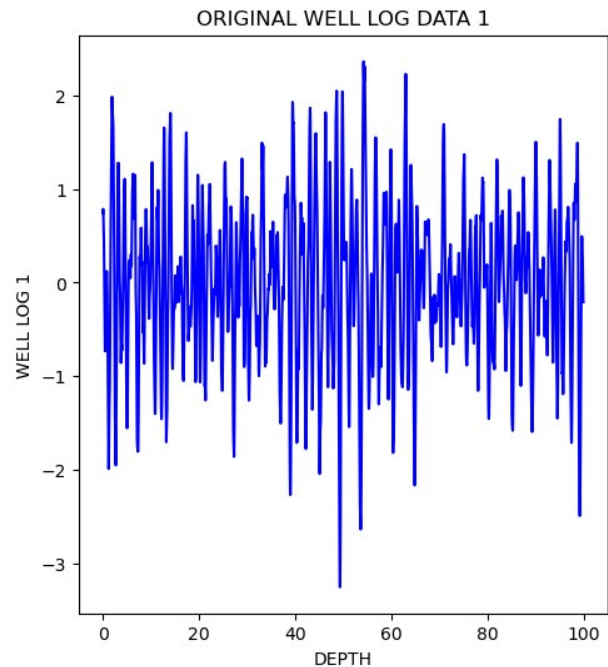
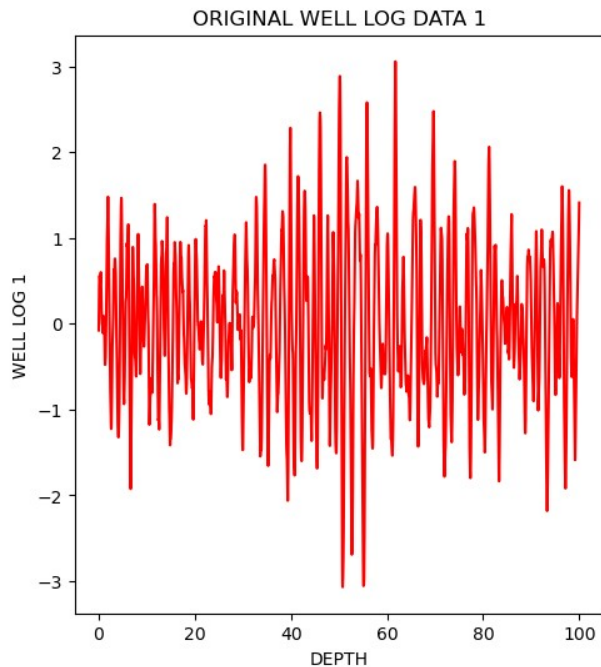
well_log_data2=df1['well_log_data2']
well_log_data2
0       0.740433
1       0.786514
2       0.751419
3       0.678940
4       0.605698
...
2994     0.141907
2995     0.018082
2996    -0.119629
2997    -0.197678
2998    -0.206680
Name: well_log_data2, Length: 2999, dtype: float64

depth=np.linspace(0,100,2999)

well_log_data1=df1['well_log_data1']
well_log_data2=df1['well_log_data2']
depth=np.linspace(0,100,2999)
plt.figure(figsize=(12,6))
plt.subplot(1,2,1)
plt.plot(depth,well_log_data1,color='red')
plt.title("ORIGINAL WELL LOG DATA 1")
plt.xlabel('DEPTH')
plt.ylabel('WELL LOG 1')
plt.subplot(1,2,2)
plt.plot(depth,well_log_data2,color='blue')

```

```
plt.title("ORIGINAL WELL LOG DATA 1")
plt.xlabel('DEPTH')
plt.ylabel('WELL LOG 1')
Text(0, 0.5, 'WELL LOG 1')
```



PATH AND DTW DISTANCE BETWEEN WELL LOGS

```
distance,path = fastdtw(well_log_data1,well_log_data2)
distance
path
print(f'DTW DISTANCE :{distance}')
print(f'ALIGNMENT PATH :{path}')
aligned_well_log_data1=[well_log_data1[i] for (i,j) in path]
aligned_well_log_data2=[well_log_data2[j] for (i,j) in path]
aligned_depth = np.linspace(0,100,len(aligned_well_log_data2))
print(aligned_depth)
```

```
DTW DISTANCE :1113.8048553466797
ALIGNMENT PATH :[(0, 0), (1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6,
5), (7, 5), (8, 5), (9, 5), (10, 5), (11, 5), (12, 5), (13, 5), (14,
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[0.00000000e+00 2.22568440e-02 4.45136880e-02 ... 9.99554863e+01
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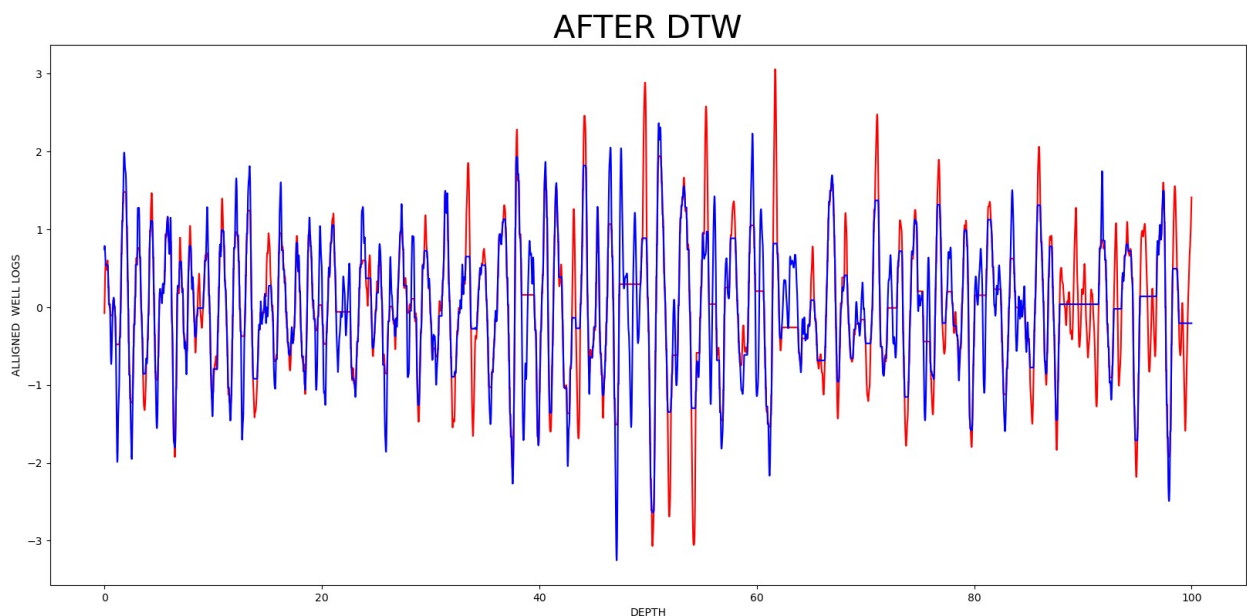
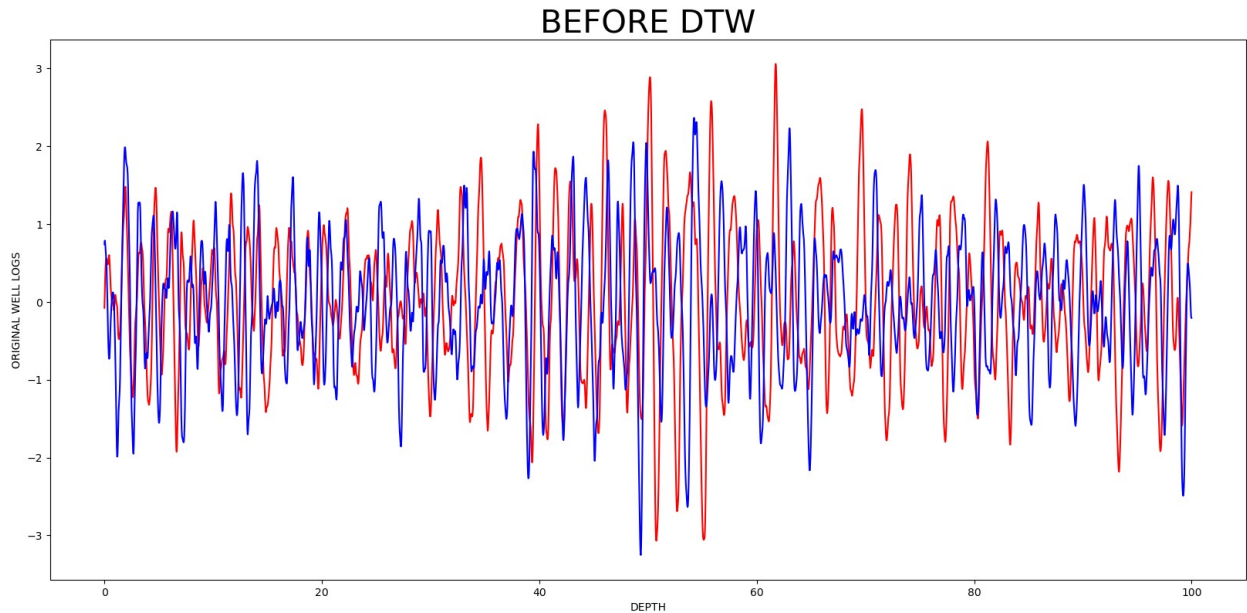
```

SEEING THE DIFFERENCE BETWEEN THE ALIGNMENT BEFORE AND AFTER APPLYING THE DTW

```

plt.figure(figsize=(20,20))
plt.subplot(2,1,1)
plt.plot(depth,well_log_data1,color='red')
plt.plot(depth,well_log_data2,color='blue')
plt.xlabel('DEPTH')
plt.ylabel('ORIGINAL WELL LOGS')
plt.title('BEFORE DTW',fontsize=30)
plt.subplot(2,1,2)
plt.plot(aligned_depth,aligned_well_log_data1,color='red')
plt.plot(aligned_depth,aligned_well_log_data2,color='blue')
plt.xlabel('DEPTH')
plt.ylabel('ALLIGNED WELL LOGS')
plt.title('AFTER DTW',fontsize=30)
plt.show()

```



APPLICATIONS OF DTW IN THE GEOPHYSICS AND IN THE OIL AND GAS INDUSTRY

```
import obspy
st = obspy.read("file_7218.sgy", format='SEGY')
print(st)
traces=st.traces
```

1983 Trace(s) in Stream:

Seq. No. in line: 0 | 2023-12-25T15:41:48.000000Z - 2023-12-

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

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```

tracel=traces[0]
tracel

Seq. No. in line:    0 | 2023-12-25T15:41:48.000000Z - 2023-12-
25T15:41:59.996000Z | 250.0 Hz, 3000 samples

time=tracel.times()
amplitude=tracel.data
print(time)
print(amplitude)

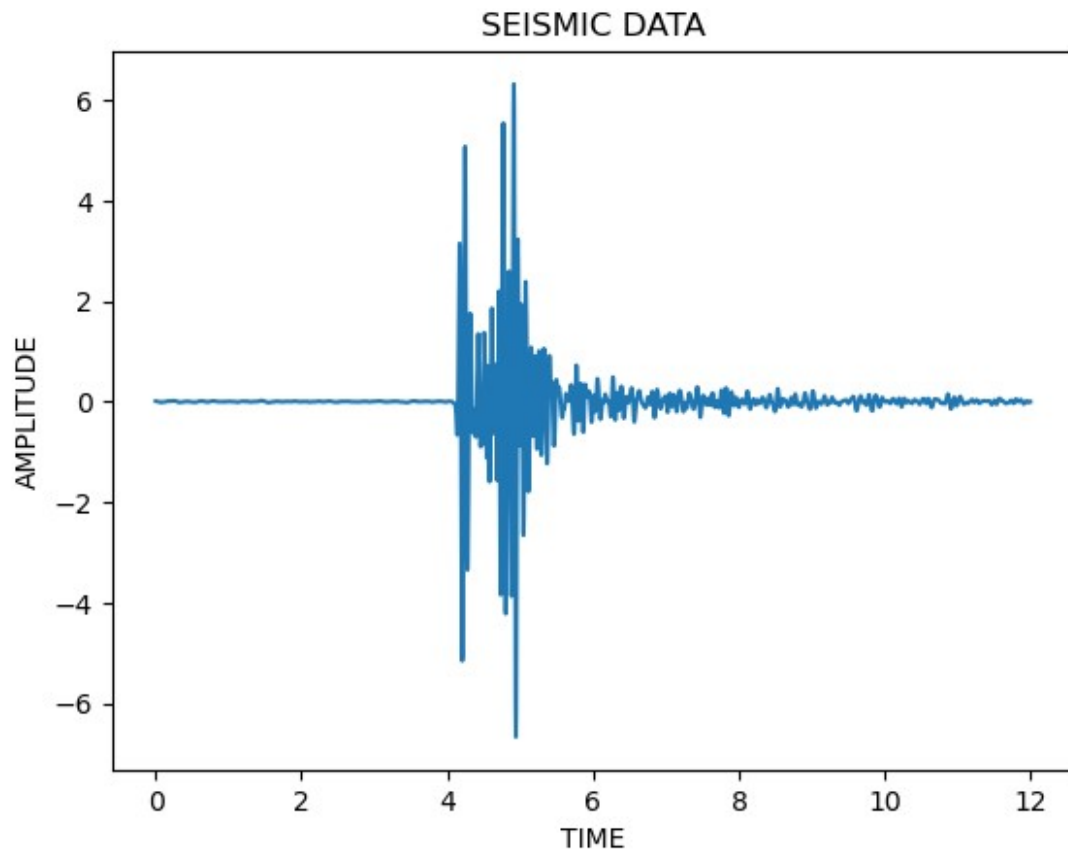
[0.0000e+00 4.0000e-03 8.0000e-03 ... 1.1988e+01 1.1992e+01
1.1996e+01]
[ 0.00867792  0.00831125  0.00611121 ... -0.00415562 -0.00220004
-0.00207781]

start_time=3.5
end_time=6
mask=(time>=start_time) & (time<=end_time)
time_selected=time[mask]
amplitude_selected=amplitude[mask]

import matplotlib.pyplot as plt
plt.plot(time, amplitude)
plt.xlabel('TIME')
plt.ylabel('AMPLITUDE')
plt.title('SEISMIC DATA')

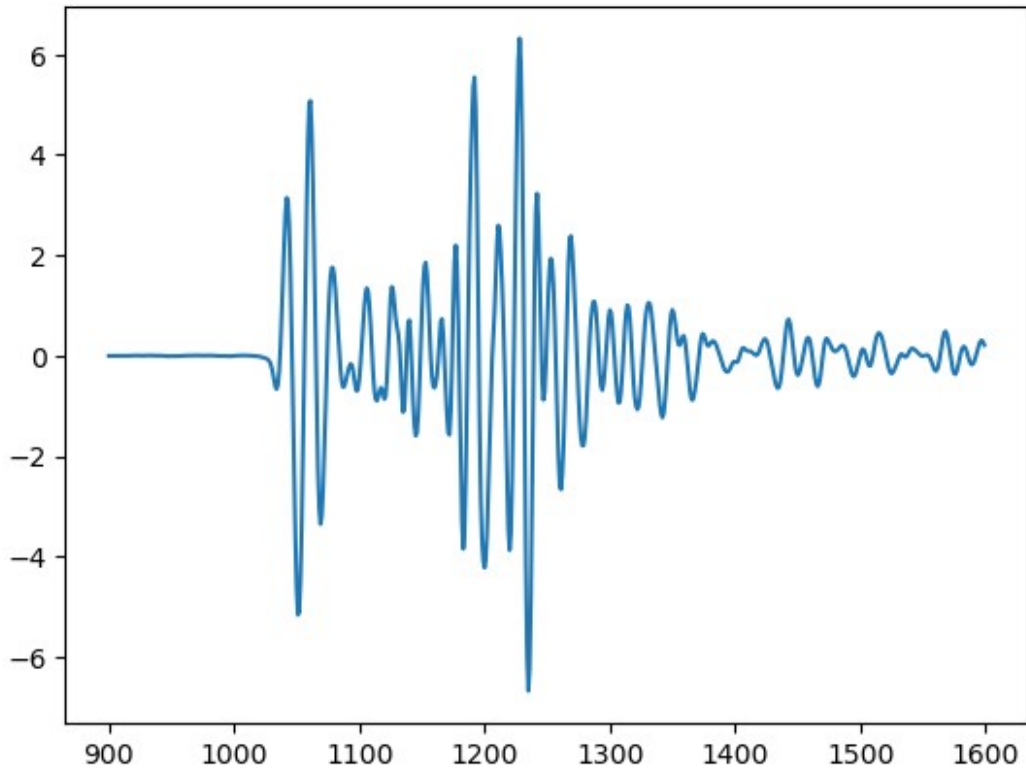
Text(0.5, 1.0, 'SEISMIC DATA')

```



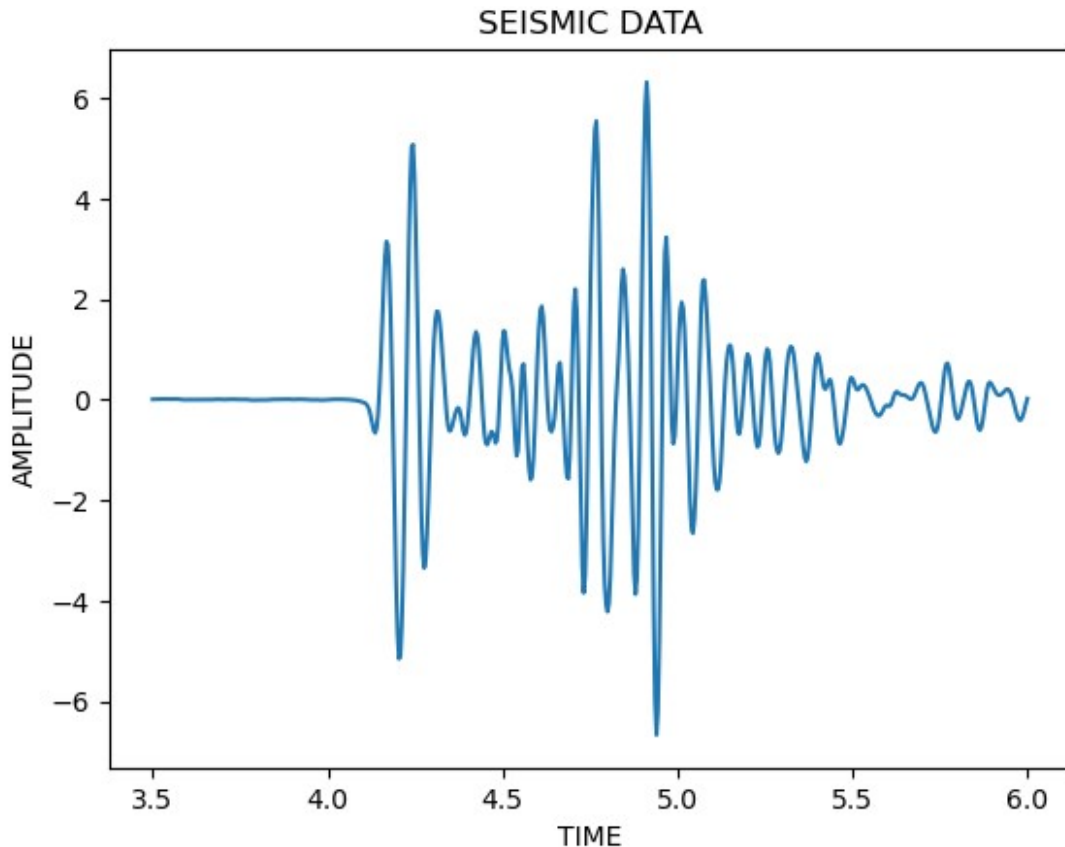
```
start,end=900,1600
trace_subset=trace1[start:end]
x_values=np.arange(start,end)
plt.plot(x_values,trace_subset)

[<matplotlib.lines.Line2D at 0x14396b5c0>]
```



```
plt.plot(time_selected, amplitude_selected)
plt.xlabel('TIME')
plt.ylabel('AMPLITUDE')
plt.title('SEISMIC DATA')
```

```
Text(0.5, 1.0, 'SEISMIC DATA')
```



```
import numpy as np
import matplotlib.pyplot as plt

def sta_lta(tracel, sta_window, lta_window, threshold):
    n_samples = len(tracel)
    sta = np.zeros(n_samples)
    lta = np.zeros(n_samples)
    sta_lta_ratio = np.zeros(n_samples)

    for i in range(n_samples):
        sta[i] = np.mean(np.abs(tracel[max(0, i - sta_window):i]))
        lta[i] = np.mean(np.abs(tracel[max(0, i - lta_window):i]))
        if lta[i] > 0:
            sta_lta_ratio[i] = sta[i] / lta[i]

    for i in range(sta_window, n_samples):
        if sta_lta_ratio[i] > threshold:
            return i
    return -1

# Example usage
sta_window = 20
lta_window = 200
```

```

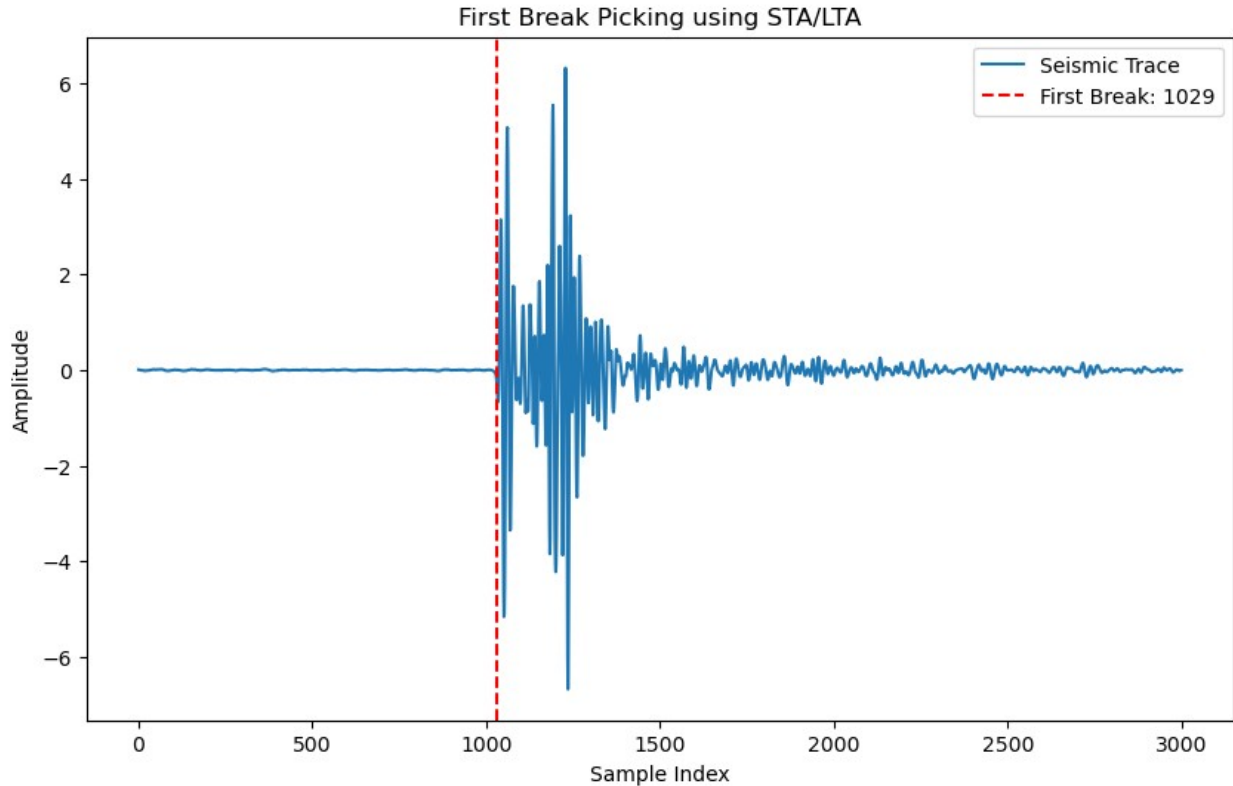
threshold = 3.0

# First break picking
fb_index = sta_lta(tracel, sta_window, lta_window, threshold)

# Visualization
plt.figure(figsize=(10, 6))
plt.plot(tracel, label="Seismic Trace")
if fb_index != -1:
    plt.axvline(fb_index, color="r", linestyle="--", label=f"First Break: {fb_index}")
plt.title("First Break Picking using STA/LTA")
plt.xlabel("Sample Index")
plt.ylabel("Amplitude")
plt.legend()
plt.show()

/opt/anaconda3/lib/python3.12/site-packages/numpy/core/_
fromnumeric.py:3504: RuntimeWarning: Mean of empty slice.
    return _methods._mean(a, axis=axis, dtype=dtype,
/opt/anaconda3/lib/python3.12/site-packages/numpy/core/_methods.py:129
: RuntimeWarning: invalid value encountered in divide
    ret = ret.dtype.type(ret / rcount)

```



```
df1=pd.read_csv('synthetic_well_logs.csv')
df1.head()
```

	Depth_Well_A (m)	Log_Well_A	Depth_Well_B (m)	Log_Well_B
0	0.000000	1.0	0.000000	1.0
1	1.010101	1.0	1.010101	1.0
2	2.020202	1.0	2.020202	1.0
3	3.030303	1.0	3.030303	1.0
4	4.040404	1.0	4.040404	1.0

```
df1.columns
```

```
Index(['Depth_Well_A (m)', 'Log_Well_A', 'Depth_Well_B (m)',  
      'Log_Well_B'], dtype='object')
```

```
import matplotlib.pyplot as plt
```

```
plt.plot(df1['Depth_Well_A (m)'],df1['Log_Well_A'])  
plt.show()  
plt.plot(df1['Depth_Well_B (m)'],df1['Log_Well_B'])  
plt.show()
```

